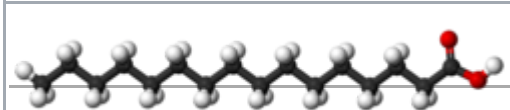
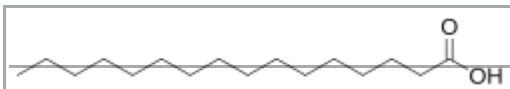




Palmitic acid

Palmitic acid^[1]



Names

Preferred IUPAC name

Hexadecanoic acid

Other names

Palmitic acid

C16:0 (Lipid numbers)

Identifiers

CAS Number

57-10-3 ([https://com
monchemistry.cas.or
g/detail?cas_rn=57-1
0-3](https://com
monchemistry.cas.or
g/detail?cas_rn=57-1
0-3)) ✓

3D model
(JSmol)

Interactive image (htt
ps://chemapps.stolaf.
edu/jmol/jmol.php?m
odel=CCCCCCCCCCC
CCCCC%28%3D
O%29O)

ChEMBL

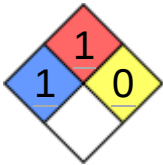
ChEMBL82293 ([http
s://www.ebi.ac.uk/che
mbldb/index.php/com
pound/inspect/ChEM
BL82293](http
s://www.ebi.ac.uk/che
mbldb/index.php/com
pound/inspect/ChEM
BL82293)) ✗

ChemSpider

960 ([https://www.che
mspider.com/Chemic](https://www.che
mspider.com/Chemic)

	al-Structure.960.htm l) ✖
ECHA InfoCard	100.000.284 (https://echa.europa.eu/substance-information/-/substanceinfo/100.000.284)
IUPHAR/BPS	1055 (http://www.guidetopharmacology.org/GRAC/LigandDisplayForward?tab=summary&ligandId=1055)
PubChem CID	985 (https://pubchem.ncbi.nlm.nih.gov/compound/985)
UNII	2V16EO95H1 (https://precision.fda.gov/unii/search/srs/unii/2V16EO95H1) ✔
CompTox Dashboard (EPA)	DTXSID2021602 (https://comptox.epa.gov/dashboard/chemical/details/DTXSID2021602)
InChI	
InChI=1S/C16H32O2/c1-2-3-4-5-6-7-8-9-10-11-12-13-14-15-16(17)18/h2-15H2,1H3,(H,17,18) ✖ Key: IPCSVZSSVZVIGE-UHFFFAOYS A-N ✖	
InChI=1/C16H32O2/c1-2-3-4-5-6-7-8-9-10-11-12-13-14-15-16(17)18/h2-15H2,1H3,(H,17,18) Key: IPCSVZSSVZVIGE-UHFFFAOYAJ	
SMILES	
CCCCCCCCCCCCCCCC(=O)O	
Properties	
Chemical formula	C ₁₆ H ₃₂ O ₂
Molar mass	256.430 g/mol
Appearance	White crystals
Density	0.852 g/cm ³ (25 °C) ^[2]

	0.8527 g/cm ³ (62 °C) ^[3]
<u>Melting point</u>	62.9 °C (145.2 °F; 336.0 K) ^[7]
<u>Boiling point</u>	351–352 °C (664–666 °F; 624–625 K) ^[8] 271.5 °C (520.7 °F; 544.6 K), 100 mmHg ^[2] 215 °C (419 °F; 488 K), 15 mmHg
<u>Solubility in water</u>	4.6 mg/L (0 °C) 7.2 mg/L (20 °C) 8.3 mg/L (30 °C) 10 mg/L (45 °C) 12 mg/L (60 °C) ^[4]
<u>Solubility</u>	Soluble in <u>amyl acetate</u> , <u>alcohol</u> , <u>CCl₄</u> , ^[4] <u>C₆H₆</u> Very soluble in <u>CHCl₃</u> ^[3]
<u>Solubility in ethanol</u>	2 g/100 mL (0 °C) 2.8 g/100 mL (10 °C) 9.2 g/100 mL (20 °C) 31.9 g/100 mL (40 °C) ^[5]
<u>Solubility in methyl acetate</u>	7.81 g/100 g ^[4]
<u>Solubility in ethyl acetate</u>	10.7 g/100 g ^[4]
<u>Vapor pressure</u>	0.051 mPa (25 °C) ^[3] 1.08 kPa (200 °C) 28.06 kPa (300 °C) ^[6]
<u>Acidity (pK_a)</u>	4.75 ^[3]
<u>Magnetic susceptibility (χ)</u>	−198.6·10 ^{−6} cm ³ /mol
<u>Refractive index (n_D)</u>	1.43 (70 °C) ^[3]
<u>Viscosity</u>	7.8 cP (70 °C) ^[3]
Thermochemistry	
<u>Heat capacity (C)</u>	463.36 J/(mol·K) ^[6]

Std molar entropy ($S_{298}^{\ominus}$)	452.37 J/(mol·K) ^[6]
Std enthalpy of formation ($\Delta_f H_{298}^{\ominus}$)	−892 kJ/mol ^[6]
Std enthalpy of combustion ($\Delta_c H_{298}^{\ominus}$)	10030.6 kJ/mol ^[3]
Hazards	
GHS labelling:	
Pictograms	 ^[2]
Signal word	Warning
Hazard statements	H319 ^[2]
Precautionary statements	P305+P351+P338 ^[2]
NFPA 704 (fire diamond)	
Flash point	206 °C (403 °F; 479 K) ^[2]
Except where otherwise noted, data are given for materials in their standard state (at 25 °C [77 °F], 100 kPa).  verify (what is  ?) Infobox references	

Palmitic acid (**hexadecanoic acid** in IUPAC nomenclature) is a fatty acid with a 16-carbon chain. It is the most common saturated fatty acid found in animals, plants and microorganisms.^{[9][10]} Its chemical formula is CH₃(CH₂)₁₄COOH, and its C:D ratio (the total number of carbon atoms to the number of carbon-carbon double bonds) is 16:0. It is a major component of palm oil from the fruit of *Elaeis guineensis* (oil palms), making up to 44% of total fats. Meats, cheeses, butter, and other dairy products also contain palmitic acid, amounting to 50–60% of total fats.^[11]

Palmitates are the salts and esters of palmitic acid. The palmitate anion is the observed form of palmitic acid at physiologic pH (7.4). Major sources of C16:0 are palm oil, palm kernel oil, coconut oil, and milk fat.^[12]

Occurrence and production

Palmitic acid was discovered by Edmond Frémy (in 1840) in the saponification of palm oil, which process remains today the primary industrial route for producing the acid.^[13] Triglycerides (fats) in palm oil are hydrolysed by high-temperature water and the resulting mixture is fractionally distilled.^[14]

Dietary sources

Palmitic acid is produced by a wide range of plants and organisms, typically at low levels. Among common foods it is present in milk, butter, cheese, and some meats, as well as cocoa butter, olive oil, soybean oil, and sunflower oil, (see table).^[15] Karukas contain 44.90% palmitic acid.^[16] The cetyl ester of palmitic acid, cetyl palmitate, occurs in spermaceti.

Palmitic acid content of common foods	
Food	% of total calories
<u>Palm oil</u>	45.1%
<u>Beef tallow</u>	26.5%
<u>Butter fat</u>	26.2%
<u>Cocoa butter</u>	25.8%
<u>Lard</u>	24.8%
<u>Cottonseed oil</u>	24.7%
<u>Chicken</u>	23.2%
<u>Corn oil</u>	12.2%
<u>Peanut oil</u>	11.6%
<u>Soybean oil</u>	11%
<u>Coconut oil</u>	8.4%
<u>Palm kernel oil</u>	8%
<u>Rapeseed oil</u>	3.6%
Source: ^[17]	

Biochemistry

Palmitic acid is the first fatty acid produced during fatty acid synthesis and is the precursor to longer fatty acids. As a consequence, palmitic acid is a major body component of animals. In humans, one analysis found it to make up 21–30% (molar) of human depot fat,^[18] and it is a major, but highly variable, lipid component of human breast milk.^[19] Palmitate negatively feeds back on acetyl-CoA carboxylase (ACC), which is responsible for converting acetyl-CoA to malonyl-CoA, which in turn is used to add to the growing acyl chain, thus preventing further palmitate generation.^[20]

Some proteins are modified by the addition of a palmitoyl group in a process known as palmitoylation. Palmitoylation is important for localisation of many membrane proteins.

Applications

Surfactant

Palmitic acid is used to produce soaps, cosmetics, and industrial mold release agents. These applications use **sodium palmitate**, which is commonly obtained by saponification of palm oil. To this end, palm oil, rendered from palm trees (species *Elaeis guineensis*), is treated with sodium hydroxide (in the form of caustic soda or lye), which causes hydrolysis of the ester groups, yielding glycerol and sodium palmitate.

Foods

Because it is inexpensive and adds texture and "mouthfeel" to processed foods (convenience food), palmitic acid and its sodium salt find wide use in foodstuffs. Sodium palmitate is permitted as a natural additive in organic products.^[21]

Military

Aluminium salts of palmitic acid and naphthenic acid were the gelling agents used with volatile petrochemicals during World War II to produce napalm. The word "napalm" is derived from the words naphthenic acid and palmitic acid.^[22]

Research

It is well accepted in the medical community that palmitic acid from dietary sources raises low-density lipoprotein (LDL) and total cholesterol.^{[17][23][24][25]} The World Health Organization have stated there is convincing evidence that palmitic acid increases cardiovascular disease risk.^[26]

A 2021 review indicated that replacing dietary palmitic acid and other saturated fatty acids with unsaturated fatty acids, such as oleic acid, could reduce several biomarkers of cardiovascular and metabolic diseases.^[27]

See also

- Retinyl palmitate
- Ascorbyl palmitate
- SN2 Palmitate
- Juniperic acid (16-hydroxypalmitic acid)

References

1. *Merck Index*, 12th Edition, **7128**.
2. Sigma-Aldrich Co., Palmitic acid (<https://www.sigmaaldrich.com/catalog/product/aldrich/258725>). Retrieved on 2014-06-02.
3. CID 985 (<https://pubchem.ncbi.nlm.nih.gov/compound/985>) from PubChem
4. "Palmitic acid" (<http://chemister.ru/Database/properties-en.php?dbid=1&id=6488>).
5. Seidell, Atherton; Linke, William F. (1952). *Solubilities of Inorganic and Organic Compounds* (<https://books.google.com/books?id=k2e5AAAAIAAJ>). Van Nostrand. Retrieved 2014-06-02.
6. n-Hexadecanoic acid (<http://webbook.nist.gov/cgi/cbook.cgi?ID=C57103&Mask=FFFF&Units=SI>) in Linstrom, Peter J.; Mallard, William G. (eds.); *NIST Chemistry WebBook* (<http://webbook.nist.gov>), *NIST Standard Reference Database Number 69*, National Institute of Standards and Technology, Gaithersburg (MD) (retrieved 2014-05-11)
7. Beare-Rogers, J.; Dieffenbacher, A.; Holm, J.V. (2001). "Lexicon of lipid nutrition (IUPAC Technical Report)" (<https://doi.org/10.1351%2Fpac200173040685>). *Pure and Applied Chemistry*. **73** (4): 685–744. doi:10.1351/pac200173040685 (<https://doi.org/10.1351%2Fpac200173040685>). S2CID 84492006 (<https://api.semanticscholar.org/CorpusID:84492006>).
8. Palmitic acid (<http://www.inchem.org/documents/icsc/icsc/eics0530.htm>) at Inchem.org
9. Gunstone, F. D., John L. Harwood, and Albert J. Dijkstra. *The Lipid Handbook*, 3rd ed. Boca Raton: CRC Press, 2007. ISBN 0849396883 | ISBN 978-0849396885
10. The most common fatty acid is the monounsaturated oleic acid. See: <https://pubchem.ncbi.nlm.nih.gov/compound/965#section=Top>
11. Gianfranca Carta; Elisabetta Murru; Sebastiano Banni; Claudia Manca (8 November 2017). "Palmitic Acid: Physiological Role, Metabolism and Nutritional Implications" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5682332>). *Frontiers in Physiology*. **8**: 902. doi:10.3389/FPHYS.2017.00902 (<https://doi.org/10.3389%2FFPHYS.2017.00902>). ISSN 1664-042X (<https://search.worldcat.org/issn/1664-042X>). PMC 5682332 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5682332>). PMID 29167646 (<https://pubmed.ncbi.nlm.nih.gov/29167646>). Wikidata Q46799280.
12. Loften, J.R.; Linn, J.G.; Drackley, J.K.; Jenkins, T.C.; Soderholm, C.G.; Kertz, A.F. (August 2014). "Invited review: Palmitic and stearic acid metabolism in lactating dairy cows" (<https://doi.org/10.3168%2Fjds.2014-7919>). *Journal of Dairy Science*. **97** (8): 4661–4674. doi:10.3168/jds.2014-7919 (<https://doi.org/10.3168%2Fjds.2014-7919>). ISSN 0022-0302 (<https://search.worldcat.org/issn/0022-0302>). PMID 24913651 (<https://pubmed.ncbi.nlm.nih.gov/24913651>).
13. Frémy, E. (1842). "Memoire sur les produits de la saponification de l'huile de palme". *Journal de Pharmacie et de Chimie*. **XII**: 757.
14. Anneken, David J.; Both, Sabine; Christoph, Ralf; Fieg, Georg; Steinberner, Udo; Westfechtel, Alfred (2006). "Fatty Acids". *Ullmann's Encyclopedia of Industrial Chemistry*. Weinheim: Wiley-VCH. doi:10.1002/14356007.a10_245.pub2 (https://doi.org/10.1002%2F14356007.a10_245.pub2). ISBN 978-3527306732.
15. "Chemical Characteristics" (<http://www.oliveoilsource.com/page/chemical-characteristics>). *Olive Oil Source*. Retrieved November 11, 2021.
16. Purwanto, Y.; Munawaroh, Esti (2010). "Etnobotani Jenis-Jenis Pandanaceae Sebagai Bahan Pangan di Indonesia" (<https://berkalahayati.org/files/journals/1/articles/265/submission/265-838-1-SM.pdf>) [Ethnobotany Types of Pandanaceae as Foodstuffs in Indonesia] (PDF). *Berkala Penelitian Hayati* (in Indonesian). **5A**: 97–108. ISSN 2337-389X (<https://search.worldcat.org/issn/2337-389X>). OCLC 981032990 (<https://search.worldcat.org/oclc/981032990>). Retrieved 10 November 2021.
17. Nelson, Gary J. (1991). *Health Effects of Dietary Fatty Acids*. American Oil Chemists' Society. pp. 84-86. ISBN 978-0935315318

18. Kingsbury, K. J.; Paul, S.; Crossley, A.; Morgan, D. M. (1961). "The fatty acid composition of human depot fat" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1205373>). *Biochemical Journal*. **78** (3): 541–550. doi:10.1042/bj0780541 (<https://doi.org/10.1042%2Fbj0780541>). PMC 1205373 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC1205373>). PMID 13756126 (<https://pubmed.ncbi.nlm.nih.gov/13756126>).
19. Jensen, RG; Hagerty, MM; McMahon, KE (June 1978). "Lipids of human milk and infant formulas: a review" (<https://doi.org/10.1093%2Fajcn%2F31.6.990>). *Am. J. Clin. Nutr.* **31** (6): 990–1016. doi:10.1093/ajcn/31.6.990 (<https://doi.org/10.1093%2Fajcn%2F31.6.990>). PMID 352132 (<https://pubmed.ncbi.nlm.nih.gov/352132>).
20. "Fatty acid biosynthesis - Reference pathway" (<http://www.genome.jp/pathway/map00061>). KEGG. Pathway Map 00061
21. US Soil Association standard 50.5.3
22. Mysels, Karol J. (1949). "Napalm. Mixture of Aluminum Disoaps" (<https://pubs.acs.org/doi/pdf/10.1021/ie50475a033>). *Industrial & Engineering Chemistry*. **41** (7): 1435–1438. doi:10.1021/ie50475a033 (<https://doi.org/10.1021%2Fie50475a033>).
23. Mensink RP, Zock PL, Kester AD, Katan MB (2003). "Effects of dietary fatty acids and carbohydrates on the ratio of serum total to HDL cholesterol and on serum lipids and apolipoproteins: a meta-analysis of 60 controlled trials" (<https://doi.org/10.1093%2Fajcn%2F77.5.1146>). *Am J Clin Nutr.* **77** (5): 1146–1155. doi:10.1093/ajcn/77.5.1146 (<https://doi.org/10.1093%2Fajcn%2F77.5.1146>). PMID 12716665 (<https://pubmed.ncbi.nlm.nih.gov/12716665>).
24. Mensink, Ronald P. (2016). "Effects of saturated fatty acids on serum lipids and lipoproteins: a systematic review and regression analysis" (<https://apps.who.int/iris/bitstream/handle/10665/246104/9789241565349-eng.pdf>). World Health Organization. Retrieved 14 March 2023.
25. Rao, Gundu HR. (2020). *Clinical Handbook of Coronary Artery Disease*. Jaypee Brothers Medical Publishers. pp. 186-187. ISBN 978-9389188301
26. "Diet, Nutrition and the Prevention of Chronic Diseases" (https://web.archive.org/web/20220901225923/https://apps.who.int/iris/bitstream/handle/10665/42665/WHO_TRS_916.pdf). World Health Organization. p. 82. Retrieved 16 March 2023.
27. Sellem, Laury; Flourakis, Matthieu; Jackson, Kim G; Joris, Peter J; Lumley, James; Lohner, Szimonetta; Mensink, Ronald P; Soedamah-Muthu, Sabita S; Lovegrove, Julie A (2021-11-25). "Impact of Replacement of Individual Dietary SFAs on Circulating Lipids and Other Biomarkers of Cardiometabolic Health: A Systematic Review and Meta-Analysis of Randomized Controlled Trials in Humans" (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9340975>). *Advances in Nutrition*. **13** (4): 1200–1225. doi:10.1093/advances/nmab143 (<https://doi.org/10.1093%2Fadvances%2Fnmab143>). ISSN 2161-8313 (<https://search.worldcat.org/issn/2161-8313>). PMC 9340975 (<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9340975>). PMID 34849532 (<https://pubmed.ncbi.nlm.nih.gov/34849532>).

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